



Tesla, the showman, in a 1889 double-exposure photograph, demonstrating high-frequency phenomena.

Nikola Tesla

A model of Tesla's first AC motor/generator patented in 1888.



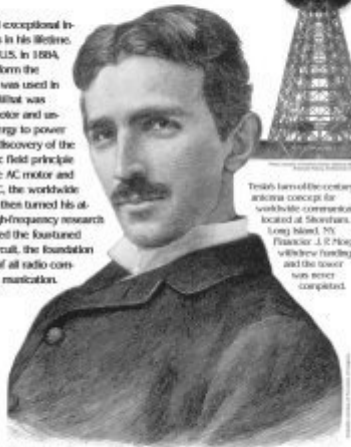
"Were we to seize and eliminate from our industrial world the results of Mr. Tesla's work, the wheels of industry would cease to turn." —B. A. Behrend

A prolific inventor of keen intelligence and exceptional insight, Tesla patented more than 500 inventions in his lifetime. He was born in Croatia and immigrated to the U.S. in 1884, bringing with him a discovery that would transform the world. By 1874, direct current (DC) technology was used in many industries, but only with minor success. What was needed was an alternating current (AC) motor and unlimited electrical energy to power industry.

Tesla's discovery of the rotating magnetic field principle brought forth the AC motor and polyphase AC, the worldwide standard. He then turned his attention to high-frequency research and discovered the four-tuned circuit, the foundation of all radio communication.



The first modern polyphase AC generating station has received the power of Niagara Falls in 1895, affording Tesla's seminal AC patents. When completely operational in 1903, it surpassed all the electrical power generated by 50 states, initiating the age of modern electrical power.



Tesla's basis of the century antenna concept for worldwide communication located at Shorham, Long Island, NY. Financier J. P. Morgan withdrew funding and the tower was never completed.

1856

July 18,
Milan, Tesla
born in
Smiljan,
Croatia

1884—Tesla files to various forms of the alternating AC patent system, and one up level to it.

1885—Tesla discovers the rotating magnetic field principle, the scientific breakthrough which produced the AC motor and the method for generating polyphase AC.

1887—The Western Union Telegraph Company helps Tesla form a business. In the next two years he produces designs for split-phase, induction, and synchronous motors and generators.

1888—Tesla secures basic patents for the polyphase AC system still in use today. He lectures to A.I.E.E. on his AC system. George Westinghouse purchases Tesla's AC patents.

1887—Tesla patents the four-tuned circuit, making radio a viable technology.

1901—Tesla patents the AC/DC converter principle, which is incorporated in all digital computer systems.

1943

Tesla dies on January 7. The U.S. Supreme Court voids the Marconi patent for the invention of radio, granting Tesla's rights for the invention. Marconi had utilized Tesla's fundamental patent No. 645,326 when he registered across the Atlantic.

1890—Tesla invents the Tesla-Grover Light Co. and makes a significant improvement in arc lighting, earning three patent awards.

1891—Tesla demonstrates his high-frequency, high-voltage Tesla coil to the A.I.E.E. He becomes a U.S. citizen.

1900—Tesla patents the VTOR, a key AC/DC converter.

1898—Tesla patents the first wireless robot, a radio-controlled model boat.

Based on the book *Nikola Tesla: The Man and His Work* by Mark Twain, 1903. © 2002 Tesla Foundation Inc. All rights reserved. For more information, visit www.teslaonline.com or www.tesla.com.